Project Title:
A Data Science Analysis of Restaurant Success in Istanbul: Effects of Location and Cuisine Type

Proposal:
In this project, my aim is to analyze the factors that influence restaurant success in Istanbul by focusing on the location (district) and cuisine type. The goal of this project is to understand whether certain combinations of location and cuisine lead to higher success.

Hypothesis: Restaurants in certain districts and cuisine types achieve higher success compared to others.

Data will primarily be collected from public sources such as Google Maps (https://www.google.com/maps) and may be complemented with additional information from platforms like TripAdvisor (https://www.tripadvisor.com). Instead of covering the entire city, I will focus on selected districts such as Kadıköy, Beşiktaş, and Şişli to ensure a more controlled and meaningful comparison. I plan to gather data on approximately 200-300 restaurants across these districts, with around 70-100 restaurants per district. The dataset will include features such as restaurant name, district, cuisine type, average rating, number of reviews, and price level. The data will be collected manually by recording information into a structured dataset. I might incorporate additional district level information such as population density or socio economic indicators from public sources. This would allow for a deeper understanding of how environmental factors influence restaurant success.

Restaurant success will be determined by using measurable indicators such as average rating and number of reviews, potentially combined into a single score (e.g., rating × log(number of reviews)) to measure success. The analysis will begin with exploratory data analysis (EDA) to identify patterns across districts and cuisine types using visualizations and summary statistics. Statistical methods such as ANOVA will be applied to test whether location and cuisine type significantly affect restaurant success. Additionally, simple predictive methods such as basic regression models will be used to understand how different factors contribute to restaurant success.